



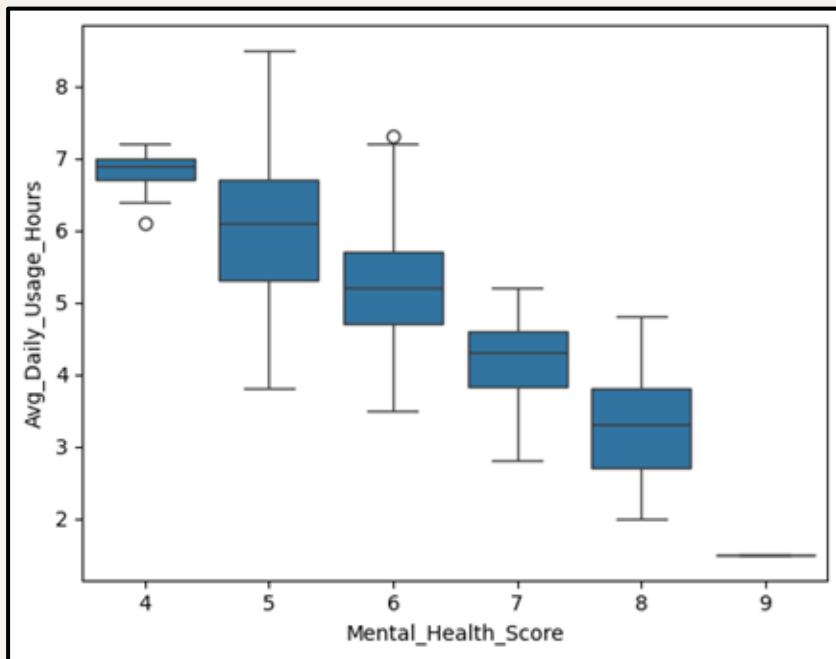
How Does Average Daily Social Media Use Affect Student Mental Health?

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The Uncontrolled Model

$$\text{Mental_Health_Score} = \beta_0 + \beta_1(\text{Avg_Daily_Usage_Hours}) + \epsilon$$



OLS Regression Results

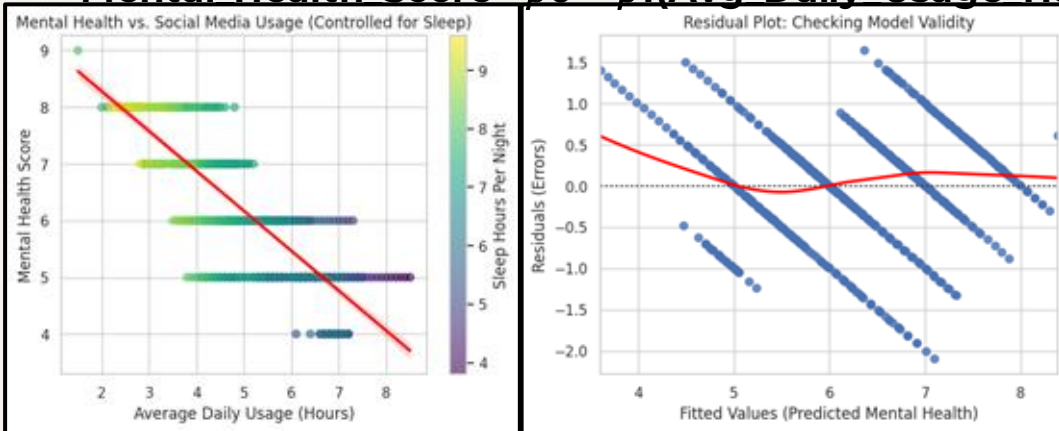
Dep. Variable:	Mental_Health_Score	R-squared:	0.642
Model:	OLS	Adj. R-squared:	0.641
Method:	Least Squares	F-statistic:	1259.
Date:	Sun, 05 Apr 2026	Prob (F-statistic):	7.85e-159
Time:	03:38:38	Log-Likelihood:	-708.48
No. Observations:	705	AIC:	1421.
Df Residuals:	703	BIC:	1430.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
Intercept	9.6898	0.101	96.199	0.000	9.492	9.888
Avg_Daily_Usage_Hours	-0.7040	0.020	-35.482	0.000	-0.743	-0.665

Omnibus: 15.754 Durbin-Watson: 2.075
Prob(Omnibus): 0.000 Jarque-Bera (JB): 16.402
Skew: -0.374 Prob(JB): 0.000274
Kurtosis: 3.010 Cond. No. 21.3

The Controlled Model

$$\text{Mental Health Score} = \beta_0 + \beta_1(\text{Avg Daily Usage Hours}) + \beta_2(\text{Sleep Hours Per Night}) + \epsilon$$



One unit increase in social media use
still decreases mental health by **0.57**.

OLS Regression Results					
Dep. Variable:	Mental_Health_Score	R-squared:	0.656		
Model:	OLS	Adj. R-squared:	0.655		
Method:	Least Squares	F-statistic:	670.4		
Date:	Wed, 22 Apr 2026	Prob (F-statistic):	1.50e-163		
Time:	15:05:12	Log-Likelihood:	-693.76		
No. Observations:	705	AIC:	1394.		
Df Residuals:	702	BIC:	1407.		
Df Model:	2				
Covariance Type: nonrobust					
	coef	std err	t	P> t	[0.025 0.975]
Intercept	7.6823	0.380	20.221	0.000	6.936 8.428
Avg_Daily_Usage_Hours	-0.5666	0.032	-17.845	0.000	-0.629 -0.504
Sleep_Hours_Per_Night	0.1939	0.035	5.472	0.000	0.124 0.263
Omnibus:	23.995	Durbin-Watson:	2.226		
Prob(Omnibus):	0.000	Jarque-Bera (JB):	25.539		
Skew:	-0.448	Prob(JB):	2.85e-06		
Kurtosis:	3.261	Cond. No.	133.		